附件：個人履歷表

附件二
Appendix 2

吳采薇

Tsai-Wei Wu

研究專長：程序系統工程、碳捕獲與再利用技術、生命週期評估、永續製程

PROFESSIONAL SUMMARY

Experienced in process systems engineering, covering the entire spectrum from conceptual design and detailed process configuration to practical control strategies. Possess over six years of hands-on industry experience in the chemical engineering sector. Currently focused on developing sustainable strategies for the chemical manufacturing industry.

SKILLS

- Process simulation: Aspen Plus, Aspen Plus Dynamic, Aspen Exchanger Design and Rating
- Control strategy design for distillation columns
- Systematic evaluation of CO_2 capture, utilization and storage technologies
- Life-cycle assessment methodologies
- Optimization, energy analysis and integration, economic and environmental analysis for plant-wide processes
- Engineering design: scale-up design, detailed diagrams, equipment and instrumentation specification, control logics and human-machine interface
- Chemical laboratory set-up and design of experiments

EDUCATION

2022年 國立台灣大學博士

Ph.D. in Chemical Engineering (2018-2022)

National Taiwan University (Process Systems Engineering Lab; Advisor: I-Lung Chien)

- Topic 1: CO2 Utilization by converting to high-value products or fuels (i.e., DMC, DEC, and DME)
- Topic 2: Design of control structure for processes with a configuration of double-column reactive-extractive distillation (i.e., resilient to disturbances by just using temperature measurements)

2014年 國立清華大學碩士 GPA: 4.14/4.30

M.S. in Chemical Engineering (2012-2014; Advisor: Chung-Sung Tan)

National Tsing Hua University (Separation and Chemical Reaction Engineering Lab)

- Topic: CO_2 Capture Using PZ/DETA Mixture from Flue Gas of Natural Gas Power Plants in a Rotating Packed Bed.

2012年 國立清華大學學士 GPA: 3.80/4.30

B.S. in Chemical Engineering (2008-2012)

National Tsing Hua University

PUBLICATIONS

I. Journal Papers

1. Miyazaki, S.; Ohno, H.; Yagihara, K.; **Wu, T. W.**; Fukushima, Y., Greenhouse Gas Emission Reduction Potential of Chemical Recycling of Polyethylene Terephthalate into Plasticizers. *ACS Sustainable Chemistry & Engineering*, **2025**, 13(37), 15509-15520. IF: 7.3
2. **Wu, T. W.;** Ohno, H.; Wang, A.; Guzman-Urbina, A.; Shinkai, Y.; Furuhashi, H.; Umezu, R.; Lin, S. T.; Fukushima, Y., Comparative analysis of CO_2-based alternative pathways for diphenyl carbonate production. *Chemical Engineering Journal*, **2025**, 520, 166354. IF: 13.4.
3. Yagihara, K.; **Wu, T. W.;** Ohno, H.; & Fukushima, Y., Evaluating the economic potential of membrane technologies for post-combustion in existing combustion systems: Comparison with MEA-based system. *Chemical Engineering Journal*, **2025**, 510, 161588. IF: 13.4; Citation: 1.
4. Wang, A.; Guzman-Urbina, A.; **Wu, T. W.**; Ohno, H; Fukushima, Y., Automated hybrid distillation sequence synthesis framework of mixed azeotropes and nonazeotropes. *Chemical Engineering Research and Design*, **2024**, 209, 26-36. IF: 3.7; Citation: 1.
5. **Wu, T. W.**; Hua, H. C., Kuo, C. T., Chien, I. L., Novel process development of hybrid reactive–extractive distillation system for the separation of a cyclohexane/isopropanol/water mixture with different feed compositions. *Industrial & Engineering Chemistry Research* **2022**, 62 (5), 2080-2089. IF: 4.2; Citation: 2.
6. **Wu, T. W.**; Chien, I. L., Novel control strategy of intensified hybrid reactive-extractive distillation process for the separation of water-containing ternary mixtures. *Separation and Purification Technology* **2022**, 294, 121159. IF: 8.6;

Citation: 15.

7. **<u>Wu, T. W.</u>**; Chien, I. L., A novel energy-efficient process of converting CO_2 to dimethyl ether with techno-economic and environmental evaluation. *Chemical Engineering Research and Design* **2022,** 177, 1-12. IF: 3.7; Citation: 17.
8. Zhang, Y. R.; **<u>Wu, T. W.</u>**; Chien, I. L., Intensified hybrid reactive-extractive distillation process for the separation of water-containing ternary mixtures. *Separation and Purification Technology* **2021,** *279*, 119712.
9. Zhang, Y. R.; **<u>Wu, T. W.</u>**; Chien, I. L., Energy-efficient heterogeneous azeotropic distillation coupling with pressure swing distillation for the separation of IPA/DIPE/Water mixture. *Journal of the Taiwan Institute of Chemical Engineers* **2022,** *130*, 103843.
10. **<u>Wu, T. W.</u>**; Chien, I. L., CO_2 Utilization Feasibility Study: Dimethyl Carbonate Direct Synthesis Process with Dehydration Reactive Distillation. *Industrial & Engineering Chemistry Research* **2020,** *59* (3), 1234-1248. IF: 4.2; Citation: 28.
11. **<u>Wu, T. W.;</u>** Hung, Y. T.; Chen, M. T.; Tan, C. S., CO_2 capture from natural gas power plants by aqueous PZ/DETA in rotating packed bed. *Separation and Purification Technology* **2017,** *186*, 309-317. IF: 8.6; Citation: 31.
12. Yu, C.-H.; **<u>Wu, T. W.</u>**; Tan, C. S., CO_2 capture by piperazine mixed with non-aqueous solvent diethylene glycol in a rotating packed bed. *International Journal of Greenhouse Gas Control* **2013,** *19*, 503-509.

II. Conference Paper

1. **<u>Wu, T. W.</u>**; Guzman-Urbina, A.; Ohno, H.; Fukushima, Y.; Liang, H. H.; Lin, S. T. *Towards Environmentally Sustainable Production of Diphenyl Carbonate: Comparative Analysis of Alternative Pathways,* Proceedings of the 34th European Symposium on Computer Aided Process Engineering, Florence, Italy, 2024.
2. **<u>Wu, T. W.</u>**; Chien, I. L. *Optimization and Heat Exchanger Network Design of Diethyl Carbonate Two-step Synthesis Process from CO_2 and Propylene Oxide*, 14th International Symposium on Process Systems Engineering (PSE 2021+), Kyoto, Japan, 2022.

III. Patents

1. **<u>Wu, T. W.</u>**; Chien, I. L.; Wang, S. J.; Wong, D. S. H.; Lee, E. K.; Jang, S. S. DEVICE AND METHOD FOR MANUFACTURING DIMETHYL CARBONATE. US 11,332,431 B2; TW I745931 ***(licenced)***, 2021.

WORK EXPERIENCE

製程工程師 – 典仲有限公司

Process Development Engineer (March 2015 – February 2023)

Reciprocal Phi Design Inc.

- Project 1/ CO_2 Capture Demo Plant/ Proprietor: Formosa Petrochemical Corporation (March 2015 – Dec. 2015)
  - Contributed experimental experiences for process design and operation
  - Designed a human-machine interface (HMI) for operating the plant
  - Started up the plant and troubleshot process problems
  - Designed SOP for starting/operating/ceasing the plant
- Project 2/ Ammonia Removal in Industrial Wastewater / Proprietor: Kuang Ming Enterprise Co. (Dec. 2014 – Dec. 2015)
  - Connected people and information between academia and industry to promote the academic technology applying to industry
  - Scaled up the ammonia stripping process using Aspen Plus simulation
  - Designed piping and instrumentation diagram (P&ID)
- Project 3/ Chemical Manufacture and Purification / Proprietor: XXX (Dec. 2016 – March 2017)
  - Scaled up from pilot-scale process to commercial scale process
  - Simulated batch reaction and distillation using Aspen Plus
  - Designed process flow diagram (PFD), P&ID
  - Designed equipment and instrumentation in commercial processes
- Project 4/ MeOH Recovery Batch Distillation Process Design/ Proprietor: HOPAX Corporation (Feb. 2017 – 2018)
  - Wrote code for sequential control logics in batch distillation process
  - Simulated batch distillation using Aspen Plus
  - Designed PFD, P&ID
- Project 5/ Solvent Recovery Process Revamping/ Proprietor: Formosa Plastics Corporation (June 2019 – June 2020)
  - Rebuilt steady-state model using Aspen Plus
  - Analyzed energy performance
  - Proposed energy-saving scenario
- Project 6/ Pressure Swing Adsorption Process Pilot Plant/ Proprietor: China Steel Corporation (March 2022 – August 2023)
  - Designed an HMI and contributed to trial run
  - Simulated PSA process with cyclic operation using Aspen Adsorption

課程助教 – 國立台灣大學

Teaching Assistant (February 2020 – June 2022)

- TA for Computer-Aided Process Design taught by Prof. I-Lung Chien
- TA for Chemical Engineering Process Design Practice taught by Prof. I-Lung Chien

博士後研究員 – 日本東北大學

Postdoctoral Researcher (March 2023 – March 2024)

Tohoku University (Environmental Green Process Study Lab). Sendai, Japan

- Industrial collaborated project (Mitsubishi Gas Company): GHG Emission Evaluation and Process Intensification for Different Routes of Diphenyl Carbonate Production
    - Research presented as an invited talk on 80th Anniversary Symposium of the Tohoku Branch Chemical Society

特任助教 – 日本東北大學

Project Assistant Professor (April 2024 – July 2025)

Tohoku University (Environmental Green Process Study Lab), Sendai, Japan

早期職涯研究者工作小組–Future Earth Taipei (2026-)

HONORS

- 國立台灣大學工學院-110學年度工學院研究生院長獎

Dean's Award of College of Engineering - "Environmental and Economic Analysis for Processes of CO_2 Conversion to Dimethyl Carbonate, Diethyl Carbonate and Dimethyl Ether", 2022, National Taiwan University

- 112年國家科學及技術委員會補助博士後赴國外研究(千里馬計畫)

2023 The Postdoctoral Research Abroad Program, sponsored by the Ministry of Science and Technology (MOST), Taiwan

- Best Oral Presentation Award

2024 Symposium on Thermodynamics and Process Systems Engineering, 3-4 May 2024. For the paper entitled "Different Routes for DPC Production: a Thermodynamic Perspective and a Practical Discussion of Process Selection".